$$X_1, X_2, \ldots, X_n \overset{iid}{\sim} F(x)$$

$$Y = X_{(1)} = \min(X_1, X_2, \ldots, X_n)$$

$$P(X_1 \le x) = F(x)$$

$$\Rightarrow 1 - P(X_1 > x) = F(x)$$

$$\Rightarrow P(X_1 > x) = 1 - F(x)$$

$$F_Y(y) = P(Y \le y)$$

$$= 1 - P(Y > y)$$

$$= 1 - P\{\min(X_1, X_2, \ldots, X_n) > y\}$$

$$= 1 - P\{X_1 > y, X_2 > y, \ldots, X_n > y\}$$

$$= 1 - P\{X_1 > y\} P\{X_2 > y\} \ldots P\{X_n > y\}$$

$$= 1 - (1 - F(y))(1 - F(y)) \ldots (1 - F(y)) \quad (\because X_1, X_2, \ldots, X_n \text{ are independent RVs})$$

$$F_Y(y) = 1 - (1 - F(y))^n$$

$$\therefore f_Y(y) = n(1 - F(y))^{n-1} f(y)$$

$$f_{X_{(1)}}(y) = n(1 - F(y))^{n-1} f(y)$$

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum(x_i-\bar{x})^2 \sum(y_i-\bar{y})^2}}$$

$$= \sum_{i=1}^{n}\left(x_i y_i - \bar{x} y_i - \bar{y} x_i + \bar{x}\bar{y}\right)$$

$$= \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y} - \cancel{n\bar{x}\bar{y}} + \cancel{n\bar{x}\bar{y}}$$

$$= \frac{S_{xy}}{(S_{xx} \cdot S_{yy})^{1/2}}$$

where $\boxed{S_{xy} = \sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})} = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}$

$$S_{xx} = \sum_{i=1}^{n}(x_i-\bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

$$S_{yy} = \sum_{i=1}^{n}(y_i-\bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2$$

$$\therefore r_{xy} = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum x_i^2 - n\bar{x}^2)(\sum y_i^2 - n\bar{y}^2)}}$$

Properties:

1. $-1 \le r \le 1$

2. If for constants a and b with $b>0$, $y_i = a + bx_i$, $i = 1, 2, \ldots, n$, then $r = 1$

If for constants a and b with $b<0$, then under the transformation $y_i = a + bx_i$, $i = 1, 2, \ldots, n$ $r = -1$

3. (x_i, y_i) $\quad$ $u_i = a + b x_i \quad v_i = c + d y_i$

r_{xy} $\quad$ $u = a + bx, \quad v = c + dy$

$$r_{uv} = \frac{bd}{|bd|} r_{xy}$$

$u_i = a + b x_i$

$\bar{u} = a + b\bar{x}$

$u_i - \bar{u} = b(x_i - \bar{x})$

$v_i - \bar{v} = d(y_i - \bar{y})$

✓ $$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$r_{uv} = \frac{\sum (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum (u_i - \bar{u})^2 \sum (v_i - \bar{v})^2}}$$

$$= \frac{\sum_{i=1}^{n} b(x_i - \bar{x}) d (y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} b^2 (x_i - \bar{x})^2 \sum_{i=1}^{n} d^2 (y_i - \bar{y})^2}}$$

$$= \frac{bd \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{|bd| \sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = \frac{bd}{|bd|} r_{xy}$$

Property 1: For all (x_i, y_i), $i = 1, 2, \ldots, n$

$$\left(\frac{x_i - \bar{x}}{\sqrt{S_{xx}}} \mp \frac{y_i - \bar{y}}{\sqrt{S_{yy}}} \right)^2 \geq 0$$

$$\Rightarrow \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{\sqrt{S_{xx}}} \mp \frac{y_i - \bar{y}}{\sqrt{S_{yy}}} \right)^2 \geq 0$$

$$\Rightarrow \sum_{i=1}^{n}\left\{\frac{(x_i-\bar{x})^2}{S_{xx}}+\frac{(y_i-\bar{y})^2}{S_{yy}}\mp 2\,\frac{(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{S_{xx}S_{yy}}}\right\}\geq 0$$

$$\Rightarrow \frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{S_{xx}}+\frac{\sum_{i=1}^{n}(y_i-\bar{y})^2}{S_{yy}}\mp 2\,\frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{S_{xx}S_{yy}}}\geq 0$$

$$\Rightarrow 1+1\mp 2r_{xy}\geq 0$$

$$\Rightarrow 2\mp 2r_{xy}\geq 0$$

$$\Rightarrow \underline{r_{xy}\leq 1}$$

$$1+r_{xy}\geq 0 \Rightarrow \underline{r_{xy}\geq -1}$$